

Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

TR-DLT-001: Load Customer and Order Data

Related Functional Requirement(s): FR-DLT-001

Objective: Implement a data ingestion pipeline to load customer and order data from CSV files to Delta tables using PySpark.

Target Cluster Configuration:

- Cluster Name: Gen AI POC Databricks Cluster
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: Standard_DS3_v2
- Worker Nodes: 2-4 Standard_DS3_v2
- Autoscaling: Enabled
- Auto Termination: 30 minutes
- Libraries Installed: delta-lake, pyspark

Source Data Details:

- Dataset Name: customerdata
- Location: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata
- Format: CSV
- Description: Customer data
- Dataset Name: orderdata
- Location: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
- Format: CSV
- Description: Order data

Target Data Details:

- Output Name: customer
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customer
- Format: Delta
- Description: Customer Delta table
- Output Name: order

- Location: gen_ai_poc_databricksco.sdlc_wizard.order
- Format: Delta
- Description: Order Delta table

Job Flow / Pipeline Stages:

- Read customer data from CSV
- Load customer data into Delta table
- Read order data from CSV
- Load order data into Delta table

Data Transformations / Business Logic:

- Step: Read CSV data
- Description: Read customer and order data from CSV files
- Transformation Logic: Use PySpark to read CSV files
- Step: Load data into Delta tables
- Description: Load customer and order data into Delta tables
- Transformation Logic: Use Delta Lake to write data into Delta tables

Error Handling and Logging:

- Log errors during data ingestion
- Handle null or malformed data

TR-DLT-002: Transform Order Data

Related Functional Requirement(s): FR-DLT-002

Objective: Implement a data transformation pipeline to add a new column "TotalAmount" to the "order" table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: order
- Location: gen_ai_poc_databricksco.sdlc_wizard.order
- Format: Delta
- Description: Order Delta table

Target Data Details:

- Output Name: order
- Location: gen_ai_poc_databricksco.sdlc_wizard.order
- Format: Delta
- Description: Order Delta table with new column "TotalAmount"

Job Flow / Pipeline Stages:

- Read order data from Delta table
- Add new column "TotalAmount"
- Write updated data back to Delta table

Data Transformations / Business Logic:

- Step: Add new column "TotalAmount"
- Description: Calculate "TotalAmount" by multiplying "PricePerUnit" and "Qty"
- Transformation Logic: Use PySpark to add new column and calculate value

Error Handling and Logging:

- Log errors during data transformation
- Handle null or malformed data

TR-DLT-003: Cleanse Customer and Order Data**Related Functional Requirement(s): FR-DLT-003**

Objective: Implement a data cleansing pipeline to remove null and duplicate records from customer and order tables.

Target Cluster Configuration: Same as TR-DLT-001**Source Data Details:**

- Dataset Name: customer
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customer
- Format: Delta
- Description: Customer Delta table
- Dataset Name: order
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.order
- Format: Delta
- Description: Order Delta table

Target Data Details:

- Output Name: customer
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customer
- Format: Delta
- Description: Cleansed customer Delta table
- Output Name: order
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.order
- Format: Delta

- Description: Cleansed order Delta table

Job Flow / Pipeline Stages:

- Read customer and order data from Delta tables
- Remove null and duplicate records
- Write cleansed data back to Delta tables

Data Transformations / Business Logic:

- Step: Remove null records
- Description: Remove rows with null values
- Transformation Logic: Use PySpark to filter out null records
- Step: Remove duplicate records
- Description: Remove duplicate rows
- Transformation Logic: Use PySpark to drop duplicates

Error Handling and Logging:

- Log errors during data cleansing
- Handle null or malformed data

TR-DLT-004: Create ordersummary Table**Related Functional Requirement(s): FR-DLT-004**

Objective: Implement a pipeline to create the "ordersummary" table if it does not exist.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details: None

Target Data Details:

- Output Name: ordersummary
- Location: gen_ai_poc_databricksco.sdlc_wizard.ordersummary
- Format: Delta
- Description: ordersummary Delta table

Job Flow / Pipeline Stages:

- Check if "ordersummary" table exists
- Create "ordersummary" table if it does not exist

Data Transformations / Business Logic: None

Error Handling and Logging:

- Log errors during table creation
- Handle exceptions during table creation

TR-DLT-005: Load ordersummary Table

Related Functional Requirement(s): FR-DLT-005

Objective: Implement a pipeline to load the "ordersummary" table with joined customer and order data.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: customer
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customer
- Format: Delta
- Description: Customer Delta table
- Dataset Name: order
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.order
- Format: Delta
- Description: Order Delta table

Target Data Details:

- Output Name: ordersummary
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.ordersummary
- Format: Delta
- Description: ordersummary Delta table with joined data

Job Flow / Pipeline Stages:

- Join customer and order data
- Load joined data into "ordersummary" table as SCD type 2

Data Transformations / Business Logic:

- Step: Join customer and order data
- Description: Join on "CustId"
- Transformation Logic: Use PySpark to join data
- Step: Load data into "ordersummary" table
- Description: Load joined data as SCD type 2
- Transformation Logic: Use Delta Lake to write data into "ordersummary" table

Error Handling and Logging:

- Log errors during data loading
- Handle null or malformed data

TR-DLT-006: Update ordersummary Table on Customer Change

Related Functional Requirement(s): FR-DLT-006

Objective: Implement a pipeline to update the "ordersummary" table when there is a change in the "customer" table.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: customer
- Location: gen_ai_poc_databrickscoe.sdmc_wizard.customer
- Format: Delta
- Description: Customer Delta table

Target Data Details:

- Output Name: ordersummary
- Location: gen_ai_poc_databrickscoe.sdmc_wizard.ordersummary
- Format: Delta
- Description: Updated ordersummary Delta table

Job Flow / Pipeline Stages:

- Detect changes in "customer" table
- Update "ordersummary" table by making old records inactive and new records active
- Update StartDate and EndDate accordingly

Data Transformations / Business Logic:

- Step: Detect changes in "customer" table
- Description: Compare current and previous versions of "customer" table
- Transformation Logic: Use PySpark to compare data
- Step: Update "ordersummary" table
- Description: Update records accordingly
- Transformation Logic: Use Delta Lake to update "ordersummary" table

Error Handling and Logging:

- Log errors during data update
- Handle null or malformed data

TR-DLT-007: Create customeragggregatespend Table

Related Functional Requirement(s): FR-DLT-007

Objective: Implement a pipeline to create the "customeragggregatespend" table if it does not exist.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details: None

Target Data Details:

- Output Name: customeraggregatespend
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customeraggregatespend
- Format: Delta
- Description: customeraggregatespend Delta table

Job Flow / Pipeline Stages:

- Check if "customeraggregatespend" table exists
- Create "customeraggregatespend" table if it does not exist

Data Transformations / Business Logic: None

Error Handling and Logging:

- Log errors during table creation
- Handle exceptions during table creation

TR-DLT-008: Load customeraggregatespend Table

Related Functional Requirement(s): FR-DLT-008

Objective: Implement a pipeline to load the "customeraggregatespend" table with aggregated data.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details:

- Dataset Name: ordersummary
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.ordersummary
- Format: Delta
- Description: ordersummary Delta table

Target Data Details:

- Output Name: customeraggregatespend
- Location: gen_ai_poc_databrickscoe.sdlc_wizard.customeraggregatespend
- Format: Delta
- Description: customeraggregatespend Delta table with aggregated data

Job Flow / Pipeline Stages:

- Aggregate "TotalAmount" from "ordersummary" table
- Load aggregated data into "customeraggregatespend" table

Data Transformations / Business Logic:

- Step: Aggregate "TotalAmount"
- Description: Aggregate by "Name" and "Date"
- Transformation Logic: Use PySpark to aggregate data

Error Handling and Logging:

- Log errors during data loading
- Handle null or malformed data

TR-DLT-009: Implement using Delta Live Tables

Related Functional Requirement(s): FR-DLT-009

Objective: Implement the data processing pipeline using Delta Live Tables.

Target Cluster Configuration: Same as TR-DLT-001

Source Data Details: Same as previous requirements

Target Data Details: Same as previous requirements

Job Flow / Pipeline Stages:

- Implement data processing pipeline using Delta Live Tables
- Cover all previous requirements

Data Transformations / Business Logic: Same as previous requirements

Error Handling and Logging:

- Log errors during data processing
- Handle null or malformed data